Amirreza Sattarzadeh Khanehbargh

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Research interests: **Natural Language Processing (NLP)** and **Machine Learning (ML)**, centred on **language models (LMs)**: tokenization and input representations, efficient adaptation and fine-tuning, and interpretability of what models encode. **Bringing NLP and ML to other fields**: **BioNLP** and **protein language models (PLMs)**; **social NLP** (hate speech); financial text; low-resource, less-studied languages and their structure (e.g. Universal Dependencies).

Education

M.Sc. in Computer Engineering Oct 2024 – expected Jul 2027
Boğaziçi University, Istanbul, Türkiye (English-taught)

- Supervisor: Prof. Arzucan Özgür, Text Analytics and Bioinformatics Lab (TabiLab) · GPA 3.25 / 4.0
- Thesis (in progress): *From Protein Sequences to Biological Language Units: Frequency- and Evolution-Aware Tokenization for Protein Language Models*
- Selected course project: Turkish NLP toolkit (UD-Turkish-BOUN)

B.Sc. in Computer Engineering Sep 2018 – Sep 2024
Iran University of Science and Technology (IUST), Tehran, Iran

- Final project: *Financial Market Trend Analysis using Sentiment Polarity in Online Text Data* (Dr. Reza Entezari-Maleki); extended into the arXiv preprint

High school diploma, Mathematics and Physics 2014 – 2018
SAMPAD (NODET: National Organization for Development of Exceptional Talents), selective gifted school, Bonab, Iran · ranked 1st among the school's Mathematics and Physics students

Research Experience

Graduate Research Assistant, LifeLU (ERC Consolidator Grant) Oct 2024 – present
TabiLab, Boğaziçi University · PI: Prof. Arzucan Özgür

- Study how tokenization interacts with adaptation in protein language models: amino-acid, BPE/Unigram and evolution-aware PUMA/evoBPE units (vocabularies of 800 to 51,200), ESM-2 models pretrained from scratch with a teammate, and frozen, full and LoRA fine-tuning.
- Built and ran a benchmark of **6,500+ training runs** of ESM-2 models on **14 PETA tasks** (function, localization, stability); analysed vocabulary-size effects, pooling heads, normalized perplexity, memory and compute cost. Found that mutation-aware PUMA units outperform frequency-based subwords in several settings and can match or surpass amino-acid tokens at certain vocabulary sizes.
- Tested fusing predicted protein-dynamics features into PLM representations. Next: interpreting the learned units, other knowledge-fusion methods and their interaction with tokenization, dynamic and task-specific tokenization.
- Regular presenter in the team's weekly reading club (PLM fine-tuning, tokenization, protein dynamics).

Developer and Researcher, PARI project Oct 2026 – present
Hrant Dink Foundation with Boğaziçi and Sabancı universities, Istanbul
Funded by the EU and the Friedrich Naumann Foundation

- Recently joined as the project's student researcher, working with its faculty, postdoctoral and PhD researchers on an open-source NLP tool for detecting hate speech and discriminatory content in Turkish and Arabic.

Team member, ELSA Health Privacy Challenge, CAMDA 2026 at ISMB 2026 Jul 2026
Organized by EMBL, CISPA Helmholtz Center and the University of Helsinki within the EU ELSA project

- Equal-contribution first author (with B. Giledereli and B. Temiz). Benchmarked single-cell RNA-seq synthetic data generators, including an LLM-based one (Cell2Sentence), on donor-split OneK1K for biological utility versus membership-inference privacy; showed that how non-HVGs are reconstructed is the main privacy lever. Selected for a talk; presented as a talk and a poster at ISMB 2026.

Research Assistant Jul 2023 – Sep 2024
Iran University of Science and Technology (IUST), Tehran · Supervisor: Dr. Reza Entezari-Maleki

- Financial NLP: labelled posts by the market move that follows them instead of human sentiment labels, and fine-tuned domain language models (FinBERT) on these labels for crypto trend prediction (up to +11% accuracy; 89.6% with market and temporal context in the prompt). Earlier: deep learning for algorithmic trading.

Publications and Posters

- **Sattarzadeh Khanehbargh, A.**, Suyunu, B., Dolu, Ö., and Özgür, A. (2026). *Tokenization-Aware Protein Language Modeling with Evolution-Guided Units and Biological Knowledge Fusion.* Poster, 25th European Conference on Computational Biology (**ECCB 2026**), Geneva. Presented in person.
- **Sattarzadeh Khanehbargh, A.** and Özgür, A. (2026). *From Protein Sequences to Biological Language Units: Frequency- and Evolution-Aware Tokenization for Protein Language Models.* Accepted poster, Lectures on Computational Linguistics (**LCS 2026**), Bolzano, Italy.
- Giledereli, B.*, Temiz, B.*, and **Sattarzadeh Khanehbargh, A.*** (2026). *Benchmarking Recent Single-Cell Synthetic Data Generation Methods on the OneK1K Dataset.* ELSA Health Privacy Challenge, **CAMDA 2026 at ISMB 2026**, Washington, DC (selected talk and poster). Preprint, arXiv:2610.31454. *Equal contribution.
- Moradi-Kamali, H., Rajabi-Ghozlou, M.-H., Ghazavi, M., Soltani, A., **Sattarzadeh, A.**, and Entezari-Maleki, R. (2025). *Market-Derived Financial Sentiment Analysis: Context-Aware Language Models for Crypto Forecasting.* Preprint, arXiv:2502.14897.

Industry Experience

Python Developer (Backend and Data) Jun 2022 – Nov 2023
Roshan-AI, Tehran (deep-learning company)

- Joined through a competitive backend internship. *Harf* (Persian speech recognition): built the data pipeline (crawling, cleaning, preprocessing of speech–text data), growing the training data in size and quality, which improved the speech models; Django APIs for model deployment. *Replai* (NLP comment/reply classification): Django backend.

Junior Software Engineer (freelance) Sep 2019 – Feb 2022
Bonab, Iran

- Telegram bot that crawls stock-market data and screens stocks; Django backend for a charity website (team of five).

Teaching and Mentoring

- **Teaching Assistant, Iran University of Science and Technology (IUST)** (Sep 2023 – Jul 2024), seven sections: Head TA, Data Science (Dr. H. Naderi); Software Engineering Mentor and Scrum Master, Systems Design and Analysis (Dr. M. Ashtiani); Scrum Master, Software Engineering II (Dr. B. Minaei-Bidgoli, Dr. M. Ashtiani); Advanced Programming, two sections (Dr. M. Maleki-Majd, Dr. M. Davoudabadi); Fundamentals of Computer Programming; Computer Workshop (Dr. M. Maleki-Majd).
- **Python instructor**, online (2023 – 2024); **volunteer instructor**, Pooyandegan-e-Daneshpazhooh-e-Mehr (BonabNGO) (2023): designed and ran a free Python and computer-engineering course for students in the region.

Skills

ML / DL	PyTorch, Hugging Face Transformers, protein language models (PLMs), fine-tuning and parameter-efficient adaptation, tokenizer training and evaluation, Weights & Biases, NumPy, pandas
NLP	Text classification (hate speech, sentiment), domain LMs (FinBERT), Turkish and Persian text processing, Universal Dependencies treebanks, speech–text data pipelines, data crawling and cleaning
Engineering	Python, Linux and remote GPU servers (SSH), Git, Django (back-end APIs), C
Languages	Persian, Azerbaijani, Turkish (native or bilingual); English (full professional, English-taught M.Sc.); Arabic (elementary)

Honors, Tests and Service

- Iran's national graduate entrance exam (2024): ranked 764th in Computer Engineering and 398th in Information Technology; chose Boğaziçi instead. National university entrance exam (2018): 233rd of ~160,000 (Mathematics). National Mathematical Olympiad, 2nd stage (2016).
- GRE Quantitative Reasoning: 165 / 170 (2024).
- Member of ISCB and AILC (Italian Association for Computational Linguistics). Board member (since 2021) of Pooyandegan-e-Daneshpazhooh-e-Mehr (BonabNGO), an educational NGO.
- Core team, Iran Innovate (Iran branch of Innovate Decentral, an international tech community), and organizing team of its event Iran Innovate (Innoventiran) (2023 – 2024).

Referees

Academic (Türkiye, Iran, UK) and industry (US, Iran) references are available on request.